1. Write the full form and one-line use of HTTP, HTTPS, FTP, SMTP.

- HTTP (Hypertext Transfer Protocol): It is used to transfer web pages and data between a web browser and a web server.
- HTTPS (Hypertext Transfer Protocol Secure): It transfers data securely between a browser and a server using encryption.
- FTP (File Transfer Protocol): It is used to transfer files between computers over a network.
- SMTP (Simple Mail Transfer Protocol): It is used to send emails from a client to a mail server or between mail servers.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

- IPv4: It uses a 32-bit address written in four decimal numbers separated by dots, such as `192.168.1.1`.
- IPv6: It uses a 128-bit address written in hexadecimal groups separated by colons, such as `2001:4860:4860::8888`. It provides a much larger number of IP addresses than IPv4.

3. Run `nslookup google.com` on your system and write down the IP address you get.

- Default Server: dns.google
- Address: 8.8.8.8

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Status Code	Meaning	Explanation
200	OK	The request was successful.
301	Moved Permanently	The requested resource has permanently moved to another URL.
404	Not Found	The requested webpage or resource could not be found.
500	Internal Server Error	The server encountered an unexpected problem.
403	Forbidden	The server understood the request but refuses to allow access.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

DNS (Domain Name System) converts human-readable domain names into IP addresses so computers can locate servers on the internet.

Example: `www.google.com`

- Root (`.`): The top of the DNS hierarchy. It directs DNS queries toward the appropriate top-level domain servers.
- TLD (`.com`): The Top-Level Domain, which comes after the final dot in a domain name.
- Domain (`google.com`): The registered domain name associated with Google.
- Subdomain (`www`): A name placed before the main domain to identify a particular hostname or service.

DNS working example: When you enter `www.google.com` into a browser, your computer uses DNS to look up the IP address associated with that hostname. The browser then uses the address to connect to the relevant server.